

Challenge (Habanero)

- Q1.** A coder mixes 2 parts of Java with 3 parts of Python to create a new programming language. What is the ratio of Java to the total mixture?
(A) 2 : 5
(B) 3 : 5
(C) 1 : 3
(D) 2 : 3
(E) 1 : 2
- Q2.** A gamer has 12 hours of gameplay data, with 3 hours of multiplayer and 9 hours of single-player. What percentage of the data is single-player?
(A) 25
(A) 50
(A) 67
(A) 75
(A) 90
- Q3.** A company mixes 4 parts of fiber optic cable with 1 part of copper wire to create a new network cable. What is the ratio of fiber optic cable to copper wire?
(A) 1 : 4
(B) 4 : 1
(C) 3 : 1
(D) 2 : 1
(E) 1 : 1
- Q4.** A data analyst has a dataset with 15 columns, 8 of which are numeric and 7 of which are categorical. What percentage of the columns are numeric?
(A) 40
(A) 50
(A) 53.33
(A) 60
(A) 66.67
- Q5.** A software developer has 18 hours of coding time, with 6 hours spent on debugging and 12 hours spent on writing new code. What is the ratio of debugging time to total coding time?
(A) 1 : 2
(B) 1 : 3
(C) 2 : 3
(D) 1 : 4
(E) 2 : 5
- Q6.** A tech company has 20 employees, 12 of whom are developers and 8 of whom are designers. What percentage of the employees are designers?
(A) 30
(A) 40
(A) 50
(A) 60
(A) 40
- Q7.** A gamer has a scoreboard with 25 points, 15 of which are from multiplayer games and 10 of which are from single-player games. What is the ratio of single-player points to total points?
(A) 2 : 5
(B) 3 : 5
(C) 1 : 3
(D) 2 : 3
(E) 3 : 8
- Q8.** A coding challenge has 30 participants, 18 of whom are beginners and 12 of whom are advanced. What percentage of the participants are advanced?
(A) 40
(A) 50
(A) 60
(A) 30
(A) 50

Open Ended Questions (FRQ)

- Q9.** A company mixes x parts of coffee with y parts of tea to create a new beverage. If the ratio of coffee to tea is 3 : 2, and the total mixture is 20 parts, find the values of x and y . Show your work and explain your reasoning.
- Q10.** A gamer has z hours of gameplay data, with p hours of multiplayer and s hours of single-player. If the ratio of multiplayer to single-player is 2 : 3, and the total gameplay time is 16 hours, find the values of p and s . Show your work and explain your reasoning.

Chef's Tips

- Check for proper units and labeling
- Ensure students provide clear explanations for their reasoning
- Remind students to show all work and calculations